

Mobile Computing

Lecture 2

Mobile IP

Why Mobile IP?

- The IP addresses are designed to work with stationary hosts because part of the address defines the network to which the host is attached.
- A host cannot change its IP address without terminating on-going sessions and restarting them after it acquires a new address.
- Other link layer mobility solutions exist but are not sufficient enough for the global Internet.

Mobility and Nomadicity

- **Mobility** is the ability of a node to change its point-of-attachment while maintaining all existing communications and using the same IP address.
- **Nomadicity** allows a node to move but it must terminate all existing communications and then can initiate new connections with a new address.

What is Mobile IP?

- **Mobile IP** is a *network layer solution* for homogenous and heterogeneous mobility on the global Internet which is scalable, robust, secure and which allows nodes to maintain all ongoing communications while moving.

Mobile IP Protocol

An Overview

- *Mobile IP routing and cellular telephone system* can be used to solve the **mobility problem**.
- However, neither is able to solve the problems related to the behavior of the wireless data link:
 - High variability in the transmission delays.
 - Occasional temporary disruptions of the transmission service.

Mobile IP Protocol

An Overview

- Mobile IP was designed to introduce mobility support for TCP/IP networking.
- Mobile IP techniques are used to manage the routing of the IP datagrams to the Mobile Node.
- This protocol uses a mediating node between the wireline world and the wireless world.
- This protocol requires no modification to existing routers or correspondent hosts.

Mobile IP Protocol

Key components

- The Mobile IP standard allows transparent routing of IP datagrams to Mobile Nodes in the Internet.
- Mobile IP protocol enables Mobile Node applications to communicate with the rest of the Internet as if it were connected to its home network.
- **Mobile Node (MN)**: the mobile device requesting access to fixed network services.
- **Home Agent (HA)**: located at the home network of the Mobile Node.
- **Foreign Agent (FA)**: located at the network to which the Mobile Node is connected.

Mobile IP Protocol

The Basics

- Each mobile node is assigned a unique **home address**.
- Hosts communicating with a mobile host are known as **correspondent hosts**.
- Correspondent hosts could be mobile or stationary.
- A correspondent host always addresses the packet to the mobile node's home address, regardless of the current location of the mobile host.
- Each mobile node must have a **home agent** on its home network.
- The home agent maintains a registry of the current location of the mobile node.
- The current location of the mobile node is identified as a **care-of address**.

Mobile IP Protocol

The Basics

- The association between a mobile node's home address and its current care-of address is called a **mobility binding**.
- The home agent should always know the current binding of each mobile node that it serves.

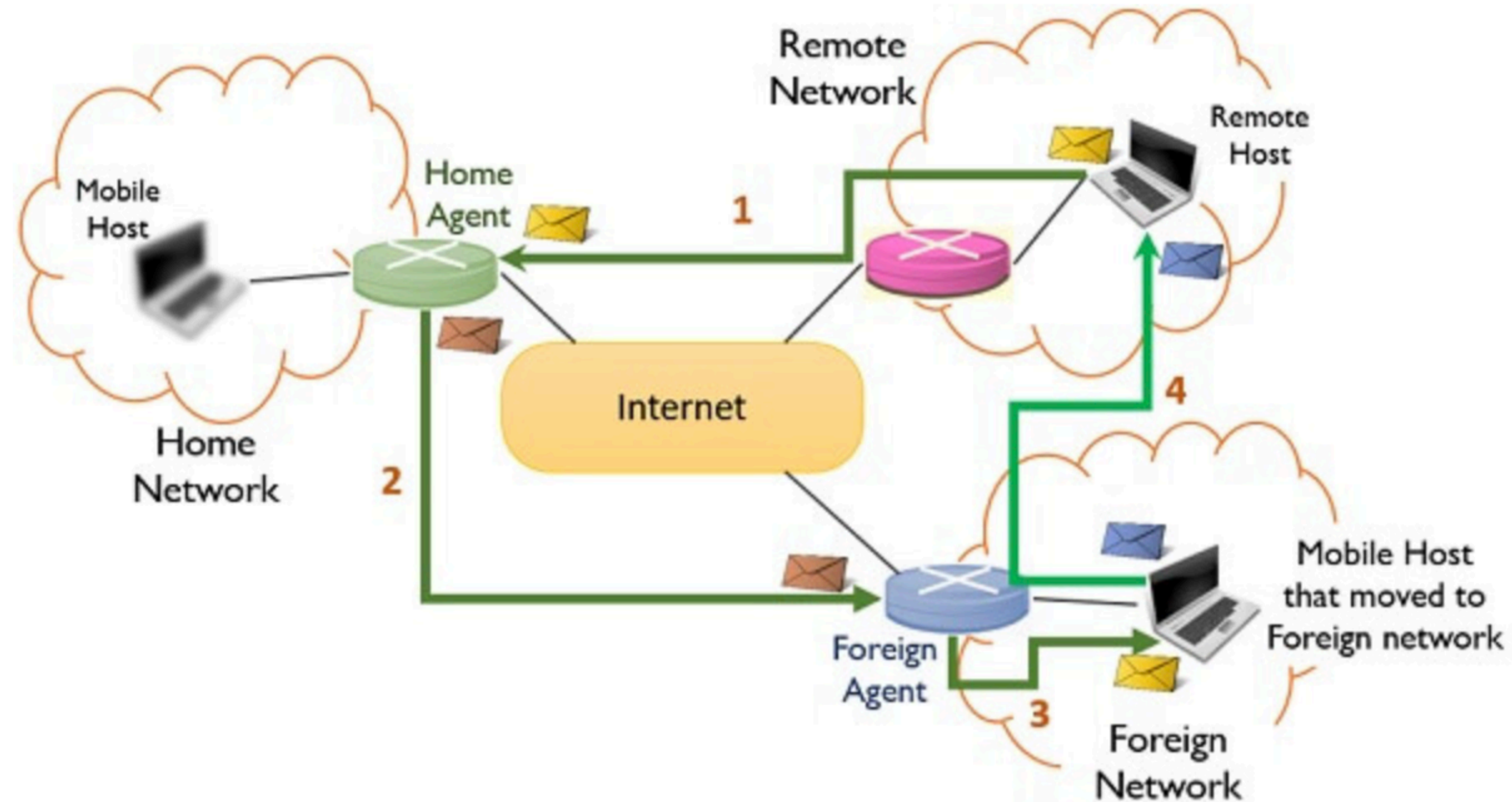
Mobile IP Protocol

How does it work

- When a Mobile Node (MN) attaches to a foreign network, it searches for a Foreign Agent (FA) within it via the *agent discovery protocol*.
- Then, the Mobile Node (MN) registers its current location with the Home Agent (HA).
- The registration is done through the Foreign Agent that forwards the registration to the Home Agent. The IP address of the FA is used as the MN's care-of address.
- After successful registration, the Home Agent intercepts all packets addressed to the Mobile Node, encapsulates them, and tunnels them to the Foreign Agent.
- The Foreign Agent decapsulates and removes any tunneling headers from the packets and delivers them to the Mobile Node.
- Outbound IP datagrams from the Mobile Node are delivered to their destination by using the standard IP routing.

Mobile IP Protocol

Mobile IP in Action



Screenshot taken from: binaryterms.com

Mobile IP Protocol

Some Implementation Details

- Home agents and foreign agents may be provided by separate nodes on a network.
- Alternatively, a single node may perform the roles of both a home agent (for its own mobile hosts) and a foreign agent (for other visiting mobile hosts).
- Another option is to use existing IP routers on the network to act as one of the agents or both.

Mobile IP Protocol

Agent Discovery

- The agent discovery protocol works as a compatible extension of the existing ICMP router discovery protocol.
- Agent discovery protocol enables a mobile node to:
 - Detect when it has moved from one network to another.
 - Detect when it has returned home.
 - Discover a suitable foreign agent in new visited networks.
- Home agents and Foreign agents periodically advertise their presence by multicasting an agent advertisement message on their networks.

Mobile IP Protocol

Agent Discovery

- Mobile nodes listen for agent advertisement messages.
- If a mobile node is receiving an advertisement message from its own home agent, it will learn that it has returned home and thus registers directly with its home agent.
- Failing to receive a number of consecutive expected advertisements, the mobile host may deduce either that it has moved or that its agent has failed.
- On detecting other advertisements, the mobile node may attempt registration with one of those foreign agents.
- A mobile node may multicast an agent solicitation message to get the attention of an agent on the connected network.

Mobile IP Protocol

Registration

- When establishing service with a new foreign agent, a mobile node must register with that foreign agent, and must also register with its home agent to inform it of its new care-of address.
- When instead establishing a new temporarily assigned local IP address as a care-of address, a mobile host must likewise register with its home agent to inform it of this new address.
- Finally, when a mobile host returns to its home network, it must register with its home agent to inform it that it is no longer using a care-of address.

Mobile IP Protocol

Registration

- To register with a foreign agent, a mobile host sends a **registration request** message to the foreign agent.
- This registration request contains the address of the MN and its HA.
- The FA forwards the request to the HA, which returns a **registration reply** message to the foreign agent.
- The FA forwards the registration reply message to the mobile host.
- When an MN registering directly with its HA, the mobile host exchanges the registration request and reply messages directly to its HA.

Mobile IP Protocol

Lifetime Period

- Each registration with an HA or FA has associated with it a **lifetime period**, negotiated during the registration.
- After this lifetime period expires, the MN's registration is deleted.
- So, to maintain continued service from the agent, the mobile host must re-register within this period.
- Note that the lifetime period may be set to infinity, in this case no re-registration is necessary.
- When a Mobile Node returns to its home network, a mobile host registers with a zero lifetime and deletes its current binding, since a mobile host needs no services of its home agent while at home.